

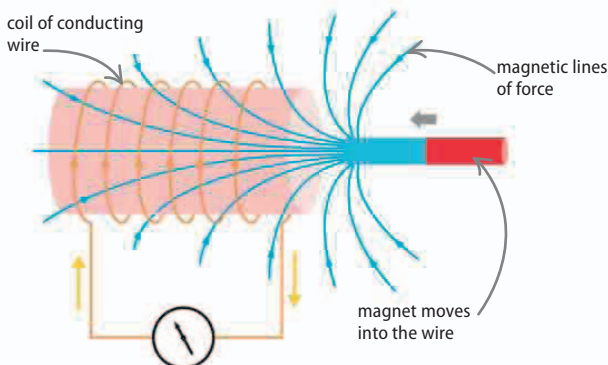
Electricity generators

GENERATORS USE INDUCTION TO CHANGE MOTION INTO ELECTRICAL POWER.

Generators, also called dynamos, are vital in many areas of technology. For example, turbines use them to change the kinetic energy of moving wind, water, or steam into electrical energy.

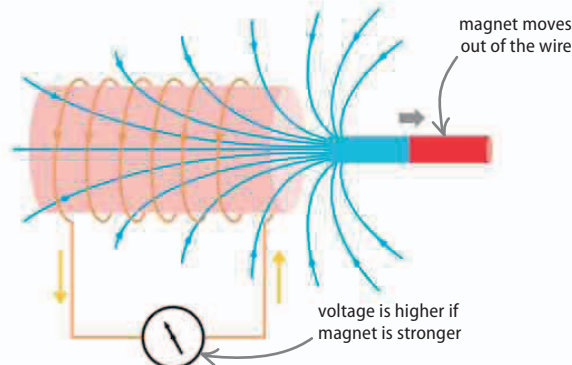
Electromagnetic induction

In 1831, English scientist Michael Faraday (1797–1867) discovered that when a magnet was moved in or out of a coil of wire, an electric current was produced. A voltage and current is produced in a conductor (the coil of wire) when it cuts across a magnetic field because the magnetic field lines of force act on the free electrons in the conductor, causing them to move. This principle, known as induction, is the basis on which all generators work.



△ Magnet moves in

A generator works when the magnet moves into the wire. The induction effect is stronger if the conductor is coiled.



△ Magnet moves out

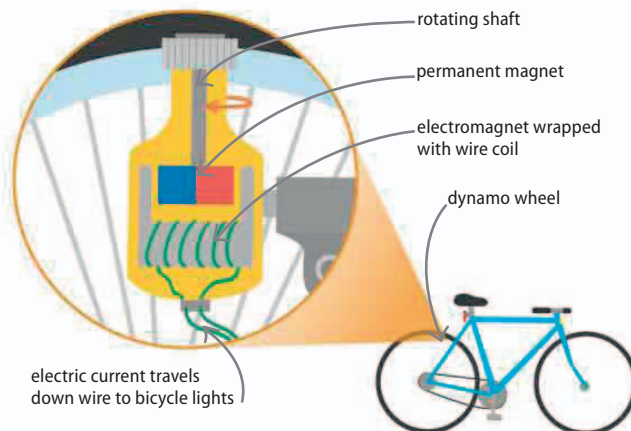
When the magnet moves out of the conductor, current is induced in the opposite direction.

Bicycle dynamo

A bicycle dynamo contains a permanent magnet fitted to a shaft. As the bicycle wheel turns, the dynamo shaft turns, rotating the permanent magnet inside a coil of wire wrapped around a soft iron core. The changing magnetic field of the turning permanent magnet induces a current in the coil, which flows from the dynamo to power the bicycle's front and rear lights.

▷ Electromagnetic induction in action

In this bicycle dynamo, the wire coil is fixed in place and the permanent magnet rotates inside it. Friction between the grooved dynamo wheel and the tire wall causes the shaft holding the magnet to rotate when the bicycle wheel turns.



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Power generation

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Built in 1871, the **Gramme dynamo** was the first electricity generator to generate power commercially.